

## BFS

```
Problems Output Debug Console Terminal Ports
• @20525pcs003-bit →/workspaces/AI-ML (main) $ python 2BFS
Route found:
central -> park -> Airport
○ @20525pcs003-bit →/workspaces/AI-ML (main) $
```

## DFS

```
Problems Output Debug Console Terminal Ports
PIN Change
• @20525pcs003-bit →/workspaces/AI-ML (main) $ python 2DFS
Main Menu
Withdraw
withdrawal
Savings
Current
Balance Check
PIN Change
○ @20525pcs003-bit →/workspaces/AI-ML (main) $
```

## A\*

```
Problems Output Debug Console Terminal Ports
• @20525pcs003-bit →/workspaces/AI-ML (main) $ python 2ASTAR

Random students:
[Student with 45 marks., Student with 54 marks., Student with 80 marks., Student with 42 marks., Student with 88 marks.]
Marks of first student:
45

Top student: Student with 88 marks.
○ @20525pcs003-bit →/workspaces/AI-ML (main) $
```

## NAÏVE BAYES

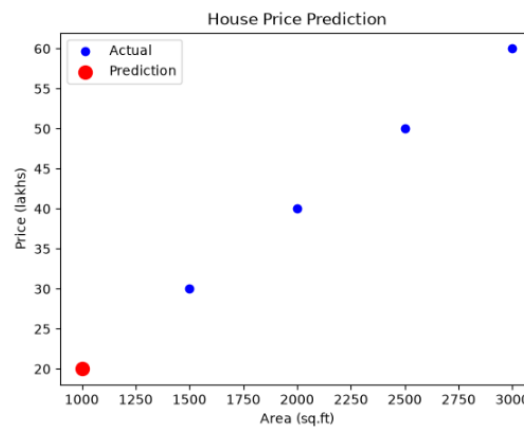
```
Problems Output Debug Console Terminal Ports
• @20525pcs003-bit →/workspaces/AI-ML (main) $ python 2NAIVE
Prediction : Don't play cricket
○ @20525pcs003-bit →/workspaces/AI-ML (main) $
```

## BAYESIAN NETWORK

```
Problems Output Debug Console Terminal Ports
@20525pcs003-bit →/workspaces/AI-ML (main) $ python 2BAYESIAN
Car Engine Analysis
-----
P(Engine fault) = 0.25
P(Noise | Engine fault) = 0.85
P(Noise) = 0.4
Probability of Engine Fault:
0.53
@20525pcs003-bit →/workspaces/AI-ML (main) $
```

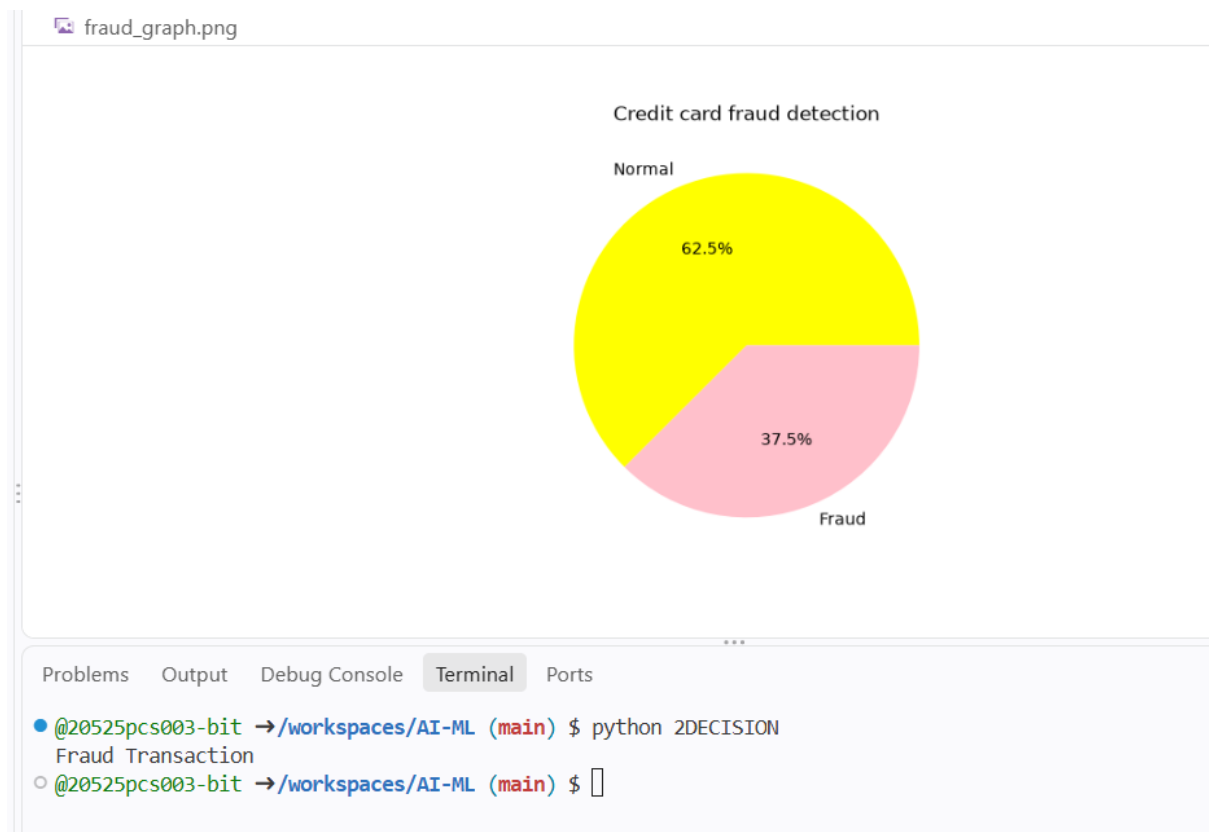
## REGRESSION MODELS

house\_price\_graph.png

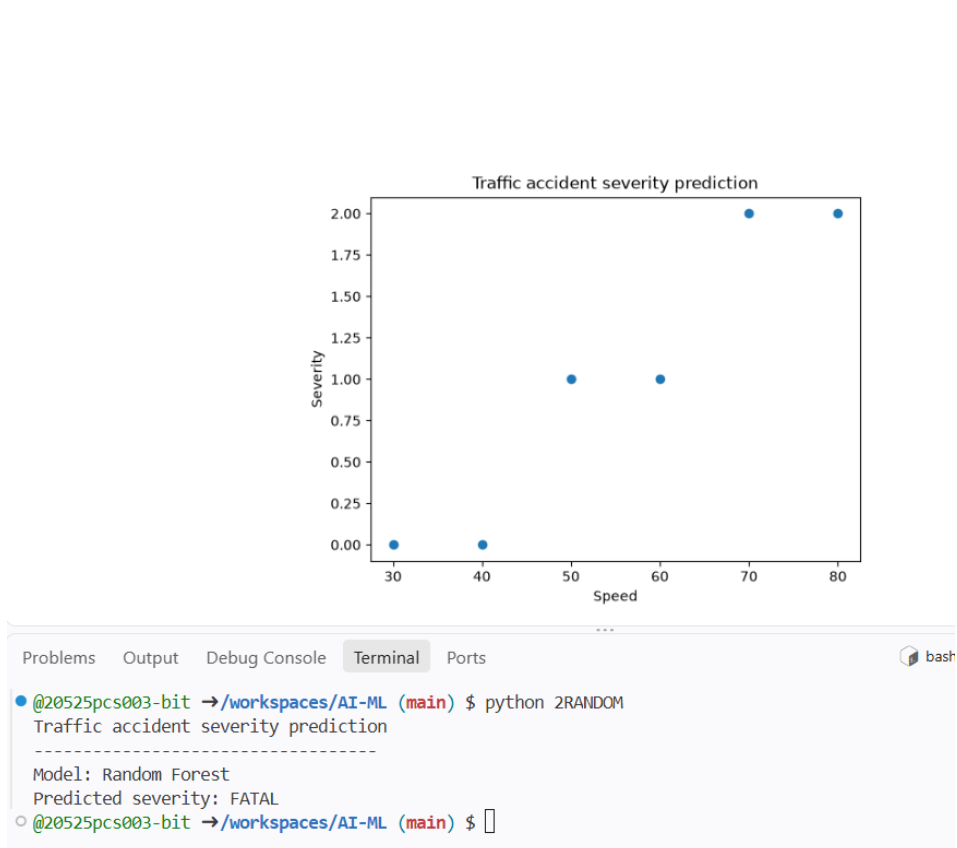


```
Problems Output Debug Console Terminal Ports
@20525pcs003-bit →/workspaces/AI-ML (main) $ python 2REGRESSION
Enter Area: 1000
/usr/local/python/3.14.2/lib/python3.14/site-packages/sklearn/utils/validation.py:2830: UserWarning:
X does not have valid feature names, but LinearRegression was fitted with feature names
  warnings.warn(
Price: ₹ 2000000.0000000005
@20525pcs003-bit →/workspaces/AI-ML (main) $
```

## DECISION TREE



## RANDOM FOREST



## SVM

```
Problems Output Debug Console Terminal Ports
@20525pcs003-bit →/workspaces/AI-ML (main) $ python 2SVM
Enter weight: 1
Enter size: 10
Predicted fruit: Orange
@20525pcs003-bit →/workspaces/AI-ML (main) $
```

## ENSEMBLING TECHNIQUES

```
Problems Output Debug Console Terminal Ports
@20525pcs003-bit →/workspaces/AI-ML (main) $ python 2ENSEMBLE
Training Data: [60, 70, 80, 90, 50]

Enter new student mark (q to stop): 95
Model 1: 95.0
Model 2: 70.0
Model 3: 82.5
Final Prediction: 82.5
Result: Pass
```

## CLUSTERING ALGORITHM

```
Problems Output Debug Console Terminal Ports python + v [ ] [ ] ... [ ]
@20525pcs003-bit →/workspaces/AI-ML (main) $ python 2CLUSTERING

Enter transaction amount (q to stop): 2000
Amount: 2000.0 -> Low Transaction

Enter transaction amount (q to stop): 8000
Amount: 8000.0 -> Low Transaction

Enter transaction amount (q to stop):
```

## EM BAYESIAN NETWORK

```
Problems Output Debug Console Terminal Ports
@20525pcs003-bit →/workspaces/AI-ML (main) $ python 2EM
Enter Traffic : 1
Traffic Jam Probability: 0.67
Prediction: Traffic Jam
@20525pcs003-bit →/workspaces/AI-ML (main) $
```

## NN MODELS

```
Problems Output Debug Console Terminal Ports
• @20525pcs003-bit →/workspaces/AI-ML (main) $ python 2NN
  Enter Distance (km): 50
  Enter Traffic Level (1-10): 7
  Prediction: 0.64
  Delivery Time: Long
○ @20525pcs003-bit →/workspaces/AI-ML (main) $
```

## DEEP LEARNING NN MODEL

```
Problems Output Debug Console Terminal Ports
• @20525pcs003-bit →/workspaces/AI-ML (main) $ python 2DEEPLARNING
  Enter Income: 50000
  Enter Credit Score: 700
  Approval Probability: 0.69
  Prediction: Loan Not Approved
○ @20525pcs003-bit →/workspaces/AI-ML (main) $
```